

Executive Summary

Python Portfolio Project

Project Overview

This project analyzes 21,663 customer interaction events across 10,000 unique sessions to understand how effectively users progress through the e-commerce customer funnel: Browse → Add to Cart → Checkout → Purchase. Using Python, Pandas, Matplotlib, Seaborn, and Plotly, the analysis evaluates funnel conversion and drop-off patterns across acquisition channels, devices, regions, product categories, time periods, bounce behavior, session duration, and customer journey paths.

Business Problem

A high volume of website traffic does not necessarily translate into purchases. The business needs to identify where customers abandon the purchasing journey, which segments perform best or worst, and which behavioral factors are associated with stronger purchase outcomes. The core problem addressed is: How can the business identify and address major sources of funnel drop-off to improve purchase conversion and revenue?

Business Questions

1. What does the overall customer funnel look like, and how many sessions reach each stage?
2. Where is the largest funnel drop-off?
3. Which acquisition channel converts browsing sessions into purchases most effectively?
4. Which acquisition channels generate the most revenue?
5. Which device has the biggest funnel performance gap?
6. Which region has the strongest and weakest purchase conversion?
7. Which product categories attract users but struggle to convert?
8. Does funnel performance vary by day of the week?
9. What hours generate the strongest purchase conversion?
10. How does bounce behavior vary across acquisition channels and devices?
11. Does session duration influence purchase conversion?
12. What are the most common customer journey paths, and where do users abandon the funnel?

Key Insights

- The overall Browse-to-Purchase conversion rate is 10.80%. Of 10,000 sessions, 7,059 reached Add to Cart, 3,524 reached Checkout, and 1,080 resulted in a Purchase.
- Checkout-to-Purchase is the largest funnel bottleneck, with a 69.35% drop-off. Add to Cart-to-Checkout also shows substantial friction, with approximately 50.08% of sessions dropping out.
- Email recorded the highest Browse-to-Purchase conversion among the analyzed channels at 11.21%, followed by Google Ads at 10.90%, Social Media at 10.86%, and Organic at 10.23%. The relatively narrow range indicates broadly similar channel conversion efficiency.
- Google Ads generated the highest channel-level revenue at approximately 312.8K, followed by Organic at 307.4K, Social Media at 280.1K, and Email at 276.1K.
- Tablet users recorded the highest overall purchase conversion at 11.52%, followed by Mobile at 11.03% and Desktop at 9.85%. Desktop performance is particularly weak at the Checkout-to-Purchase stage.
- South was the strongest-performing region at 11.25% purchase conversion, while West was lowest at 10.38%.
- Electronics had the highest product-category purchase conversion at 11.19%, while Beauty was lowest at 10.16%. Sports had the strongest Add-to-Cart rate at 71.85%, indicating strong initial engagement.
- Conversion varied by day of week: Monday was highest at 11.95%, while Tuesday was lowest at 9.41%.
- Entry-hour conversion ranged from 8.31% at 11:00 to 12.88% at 17:00, indicating variation in purchase intent by session entry time.
- Overall bounce rate was 89.20%. Channel bounce rates were tightly grouped, from 88.81% for Organic to 89.80% for Email, suggesting channel alone does not explain the high bounce level.
- Session duration showed a strong association with conversion in this dataset: 0–1 minute and 1–5 minute sessions recorded no purchases, while 5–10 minute sessions converted at 20.93% and 10–20 minute sessions at 100%. This is an observed pattern, not proof that longer sessions cause purchases.
- The most common journey paths included Browse → Add to Cart (3,535 sessions) and Browse-only (2,941 sessions), reinforcing that substantial abandonment occurs before users reach the later funnel stages.

Recommendations

- Prioritize Checkout-to-Purchase optimization by investigating payment failures, checkout complexity, unexpected costs, trust barriers, and page performance.
- Investigate the exceptionally high bounce rate through landing-page relevance, page speed, acquisition-message alignment, and targeting quality.
- Review the desktop checkout experience because Desktop has the lowest overall conversion and particularly weak final-stage performance.
- Allocate marketing effort using both conversion efficiency and revenue contribution rather than traffic volume alone; Google Ads is the leading revenue contributor while Email shows strong conversion efficiency.
- Investigate regional performance differences, particularly the gap between South and West, including customer preferences, campaign effectiveness, product availability, and user experience.
- Examine product categories with strong engagement but weaker purchase conversion for pricing, product information, availability, and checkout friction.
- Use high-conversion days and entry-hour patterns to inform campaign timing, promotional activity, and operational planning.
- Use product discovery, recommendations, relevant content, and clear calls-to-action to encourage deeper engagement, while avoiding the assumption that longer sessions themselves cause purchases.

Conclusion

The analysis shows that the primary challenge is not generating customer traffic, but converting existing engagement into completed purchases. Although 70.59% of sessions progress from Browse to Add to Cart, only 35.24% reach Checkout and 10.80% ultimately complete a purchase. The largest opportunity is therefore in the lower funnel, particularly Checkout-to-Purchase conversion, alongside the need to address the exceptionally high bounce rate. Segment-level analysis adds further opportunities across devices, regions, products, channels, and time periods. Overall, the project provides a data-driven framework for prioritizing conversion optimization, checkout improvements, customer-experience enhancements, and more effective marketing allocation.